THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE FULL SYLLABUS PAPER SET 06

Full Syllabus (Chapters 1–13) | RBSE Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. The speed of light is denoted by ‘c’ because: 1

(a) c stands for current
(b) it was chosen at random by a school
(c) c stands for centimetre
(d) it comes from the Latin word celeritas meaning speed

2. In science, a ‘theory’: 1

(a) is the same as a law
(b) cannot change with new evidence
(c) provides a tested explanation of why patterns occur
(d) means an untested guess

3. Ethanol and water are miscible. Their mixture is: 1

(a) a suspension **(b)** a colloid **(c)** heterogeneous **(d)** homogeneous

4. Mustard oil and water can be separated using: 1

(a) a magnet **(b)** a chromatography strip
(c) a condenser **(d)** a separating funnel

5. In the formation of NaCl, sodium: 1

(a) loses one electron to form Na^+ **(b)** loses two electrons
(c) gains one electron to form Na^- **(d)** shares one electron

6. The molecular/formula unit mass of sulfuric acid, H_2SO_4 is (H = 1, C = 12, N = 14, O = 16, Na = 23, Mg = 24, S = 32, Cl = 35.5, Ca = 40): 1

(a) 196 u **(b)** 100 u **(c)** 96 u **(d)** 98 u

7. The atomic number of an element is equal to the number of: 1

(a) nucleons **(b)** neutrons
(c) valence electrons **(d)** protons in its nucleus

8. A runner completes one full lap of a 400 m track. Her displacement is: 1

(a) 400 m **(b)** 800 m **(c)** 200 m **(d)** zero

9. A body starts with velocity 0 m s^{-1} and accelerates uniformly at 2 m s^{-2} for 5 s. The distance travelled is: 1

(a) 50 m **(b)** 25 m **(c)** 12.5 m **(d)** 10 m

10. Two forces of 15 N and 9 N act on a block along the same straight line in opposite directions. The magnitude of the net force is: 1

(a) 6 N **(b)** 15 N **(c)** 135 N **(d)** 24 N

11. Friction acting on a moving object is directed: 1

(a) downwards **(b)** perpendicular to the surface only
(c) along its motion **(d)** opposite to its motion

12. The SI unit of work is: 1

(a) pascal **(b)** joule **(c)** newton **(d)** watt

13. An echo is heard 0.5 s after a shout. If the speed of sound is 340 m s^{-1}, the distance of the reflecting surface is: 1

(a) 85 m **(b)** 170 m **(c)** 42.5 m **(d)** 95 m

14. The 'powerhouse of the cell' is the: 1

(a) ribosome **(b)** lysosome **(c)** Golgi apparatus **(d)** mitochondrion

15. The statement 'new cells arise only from pre-existing cells' was given by: 1

(a) Rudolf Virchow **(b)** Schwann **(c)** Brown **(d)** Hooke

16. The tissue made of dead cells with lignin-thickened walls is: 1

(a) sclerenchyma **(b)** parenchyma
(c) collenchyma **(d)** phloem parenchyma

17. The science of grouping organisms into categories is: 1

(a) reproduction **(b)** photosynthesis **(c)** evolution **(d)** classification

18. The sex of a human baby is determined by the chromosome from the: 1

(a) environment **(b)** mother **(c)** father (X or Y) **(d)** doctor

19. After fertilisation, the ovule develops into a: 1

(a) pollen grain **(b)** stigma **(c)** seed **(d)** fruit

20. The maximum insolation at the surface under a clear sky is about: 1

(a) 1 kW m^{-2} **(b)** 10 kW m^{-2} **(c)** 1.4 kW m^{-2} **(d)** 0.1 kW m^{-2}

SECTION — B (Very Short Answer Type) 8 × 2 = 16

Answer in about 10–15 words each.

21. Why are weather forecasts usually reliable for a few days but not for much longer? 2

22. Name the compounds: 2

(a) CO
(b) N_2O_5
(c) H_2S
(d) SF_6.

23. Write the three kinematic equations for uniformly accelerated motion. 2

24. What are balanced forces? Give an example. 2

25. Why do mature human RBCs lack a nucleus? State one consequence. 2

26. Name the three types of muscle tissue. 2

27. Name the male and female reproductive gametes in humans. 2

28. Name two ways in which an individual can reduce greenhouse gas emissions. 2

SECTION — C (Short Answer Type) 8 × 3 = 24

Answer in about 30–40 words each.

29. State three properties of colloids. Give two examples. 3

30. Chlorine has two isotopes of mass 35 u and 37 u in the ratio 3 : 1. Calculate its average atomic mass. 3

31. State two kinds of energy transfer other than doing work. 3

32. Describe how a microphone and a speaker work. 3

33. Name the types of epithelial tissue with one example each. 3

34. Describe the taxonomic hierarchy with an example. 3

35. Distinguish between ionic and covalent compounds on the basis of 3
(i) bonding
(ii) melting point
(iii) conduction of electricity.

OR

Why do noble gases exist as single atoms? Give two examples. 3

36. Describe the carbon cycle in brief. 3

OR

Suggest three actions to restore balance among the Earth's spheres. 3

SECTION — D (Long Answer Type) 4 × 5 = 20

Answer in about 100–150 words each.

37. (i) Explain the formation of an ionic bond in NaCl. 5
(ii) Name the ions and write their electronic configurations.
(iii) Write the formula of magnesium oxide and explain how it forms.

OR

(i) State the Law of Conservation of Mass. 5
(ii) Describe an experiment to verify it using sodium sulfate and barium chloride.
(iii) Why is a closed system required if gas is produced?

38. (i) Why do airbags and cricket fielders reduce force on impact? 5
(ii) A 0.15 kg ball at 30 m s^{-1} is stopped in 0.1 s; find the force.
(iii) Find the force if stopped in 0.01 s.

OR

(i) State Newton's first law. 5
(ii) Describe Galileo's reasoning.
(iii) Give two everyday examples of inertia.

39. **Case study:** Cell X swells and cell Y shrinks when placed in two solutions. 5
(i) What is the direction of water movement in each?
(ii) Which solution is more concentrated?
(iii) If X is a plant cell, why does it not burst?
(iv) What term describes the shrinking of the inner content of a plant cell like Y?

OR

INTERMIND

(i) Describe the structure of the cell membrane. 5
(ii) How does it control the passage of materials?
(iii) Distinguish diffusion and osmosis.

40. (i) Draw and label a flower. 5
(ii) State the function of each part.
(iii) Distinguish unisexual from bisexual flowers.

OR

Case study: A farmer plants two varieties of maize side by side and finds seeds form only when pollen reaches the other variety. 5
(i) Name the process.
(ii) Which agent likely carries pollen?
(iii) Why is it advantageous?
(iv) Name the male and female parts involved.

END OF QUESTION PAPER **Total: 80 Marks**

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